

Real data manipulation



Figure 1: License: CC BY-SA 4.0

Lab – Geospatial Analysis of Irish Climate Stations with OpenSearch

Context

link provides :

- A dataset of Irish meteorological stations (*geojson format*)
 - geographic coordinates (lat/lon)
 - elevation (m)
- Monthly aggregated observations per station, including (*zip files*):
 - precipitation (mm)
 - mean temperature (°C)

Your goal is to design a geospatial analytical system in OpenSearch that enables:

- spatial queries
- statistical aggregations
- parent/child relationships
- cartographic visualization
- climate similarity analysis (k-NN)

Learning Objectives

By the end of this lab, you should be able to:

- Design a proper OpenSearch mapping for geospatial data
- Implement a parent/child relationship
- Understand and use routing
- Perform spatial queries
- Build analytical aggregations
- Create interactive dashboards
- Implement vector-based similarity search

Part 1 – Data Modeling & Index Creation

Create an index named: `camp_stations`

It must include the following fields:

- Station fields (parent)
 - stno (keyword)
 - name (keyword)
 - county (keyword)
 - elevation (float)
 - coordinates (geo_point)
 - location (text)
 - open_date (date)
 - river (keyword)
- Observation fields (child)
 - time (date)
 - rain_mean (float)
 - temp_mean (float)
 - pluvio_mean (float)
 -

Relationship field

A join field named `station_to_observation`

- Parent: station
- Child: observation

Questions

- Why should county be mapped as keyword rather than text?
- Why is geo_point required for spatial analysis?
- What is the purpose of routing in parent/child relationships?

Part 2 – Data Indexing - Bulk Import

You must index:

- All stations (parent documents)
- All monthly observations (child documents)

Constraints

- The station `_id` must equal stno
- Child documents must use routing equal to stno
- The child must reference its parent in the relation field

Validation Tasks

- Count the number of stations
- Count the number of observations
- Test a `has_child` query

Part 3 – Attribute Queries & Aggregations

- Find stations located above 200 meters elevation
- Compute the average annual rainfall per station
- Determine the wettest month nationwide (on average)

Conceptual Questions

- What is the difference between a terms aggregation and a histogram aggregation?
- Why are aggregations efficient in OpenSearch?

Part 4 – Geospatial Queries

- Find stations within 100 km of Dublin
- Retrieve stations inside a given bounding box
- Create a spatial aggregation using `geotile_grid`

Conceptual Questions

- How does OpenSearch internally index geographic coordinates?
- What is the difference between a search engine spatial index and a classical GIS R-tree index?

Part 5 – Parent/Child Queries

- Find stations that had at least one month with rainfall > 200 mm (Use `has_child`)
- Retrieve all monthly observations for a given station (Use `has_parent`)

Discussion

Compare:

- SQL relational joins
- OpenSearch document-based joins
- Discuss advantages and limitations.

Part 6 - Dashboard Construction

Using OpenSearch Dashboards, build an analytical dashboard including:

- A map displaying stations
- Symbol size proportional to average annual rainfall
- A temperature histogram
- A county filter
- A month/year filter

Expected Outcome

An interactive decision-support cartographic interface.

Part 7 – Climate Similarity (Advanced / Bonus)

Create a field:

```
climate_signature = [annual_rain_mean, annual_temp_mean]
```

Map it as a knn_vector.

Exercise

Find the 5 stations most similar to a selected station based on: * rainfall * temperature

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